Ex. No: 9

Title: Multilayer Perceptron Implementation

Packages: TensorFlow, Keras, NumPy, Matplotlib

AIM:

To build and train a Multilayer Perceptron (MLP) using TensorFlow and Keras for handwritten digit classification.

ALGORITHM:

1. Start the program and import the required libraries.
2. Load the digits dataset and scale the pixel values to 0-1.
3. Split the data into training and testing sets.
4. Build the MLP model with two hidden Dense layers (ReLU) and one output layer (Softmax).
5. Compile the model using the Adam optimizer and sparse categorical cross-entropy loss.
6. Train the model for 20 epochs.
7. Evaluate the model on the test data and print the accuracy.
8. Plot the training and validation accuracy and stop.

PROGRAM:

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from tensorflow import keras
from tensorflow.keras import layers

# Load digits dataset (8x8 images of digits 0-9)
X, y = load_digits(return_X_y=True)
X = X / 16.0
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)

# Multilayer Perceptron
model = keras.Sequential([
    layers.Input(shape=(64,)),
    layers.Dense(64, activation="relu"),
    layers.Dense(32, activation="relu"),
    layers.Dense(10, activation="softmax")
])
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy",
metrics=["accuracy"])
model.summary()

history = model.fit(X_train, y_train, epochs=20, validation_split=0.2, verbose=0)
loss, acc = model.evaluate(X_test, y_test, verbose=0)
print("Test Accuracy =", round(acc, 4))

plt.plot(history.history["accuracy"], label="Train Accuracy")
plt.plot(history.history["val_accuracy"], label="Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("MLP Training")
plt.legend()
```

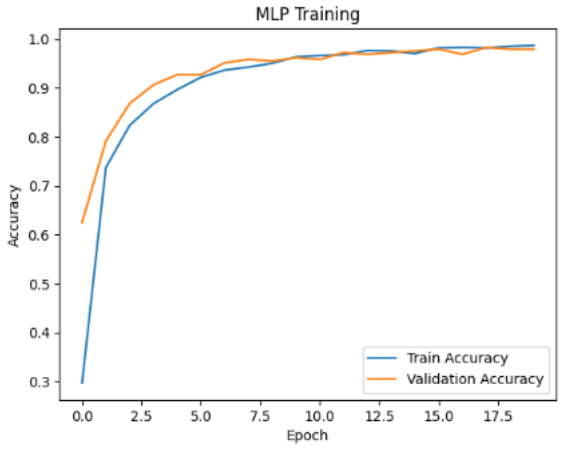
plt.show()

OUTPUT:

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	4,160
dense_1 (Dense)	(None, 32)	2,080
dense_2 (Dense)	(None, 10)	330

Total params: 6,570 (25.66 KB)
Trainable params: 6,570 (25.66 KB)
Non-trainable params: 0 (0.00 B)
Test Accuracy = 0.9667



RESULT:

The Multilayer Perceptron was built and trained successfully using TensorFlow and Keras, and it classified the digits with high test accuracy.